Init Process

Friday, September 19, 2025 9:39 AM

## Busybox

└→ init → /sbin/init → /etc/

inittab

device : : action : Command

- Sysinit
- wait
- askfirst

mount :

- procfs → RAM
- devtmpfs → RAM
- sysfs → RAM
- tmpfs → sdcard

Kernel

Process

← mirror for user space to know what kernel provide

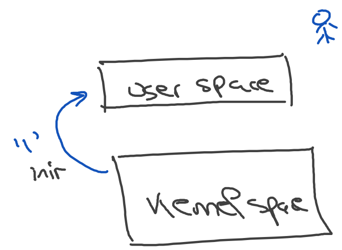
reboot

clean

## What Init process do:

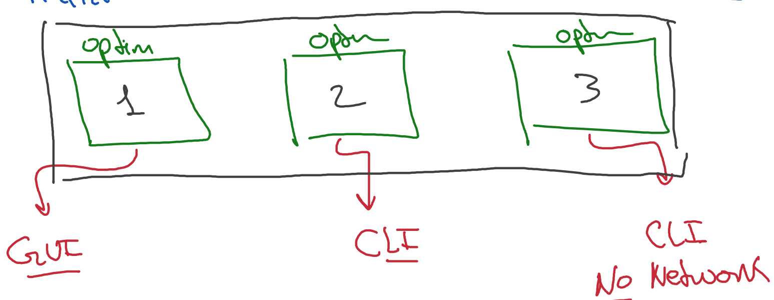
1. mount pseudo filesystem
2. Initialize env variable
3. Initialize GUI
4. Initialize modules

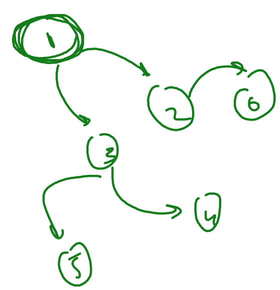
} Initialize user environment




## Runlevels (feature)

Initialize user environment in different manners




Linux standard

0 → 6

7 options

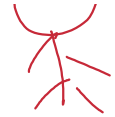
| Run Level | Mode | Action |
| --- | --- | --- |
| 0 | Halt | Shuts down system Close the Syste |
| | | Does not configure network interfaces, start |

| | | |
|---|---|---|
| 1 | Single-User Mode | Does not configure network interfaces, start daemons, or allow non-root logins Root ← recovery |
| 2 | Multi-User Mode | Does not configure network interfaces or start daemons. CLI |
| 3 | Multi-User Mode with Networking | Starts the system normally. CLI |
| 4 | Undefined | Not used/User-definable |
| 5 | X11 | As runlevel 3 + display manager(X) |
| 6 | Reboot | Reboots the system Restart |

cpu load
Power →

cpu
load →

← default
Start GUI



Cybersecurity

Linux
③
⑤ fady
python
cpu
~~GUI~~

Linux
Emad ⑤
↳ cpu load
↳ ethernet
X → discard frame

GUI
20% 50%
80%
90%

Embedded Linux

Option 1 → User → QT, Net, models

Option 2 → maintenance config → OBD, Bootloader

Why during flashing
to keep the QT function running?

System V
↳ device : runlevels : action : Command
↳ used & Introduced By UNIX

⑤ System
SystemV → option 3, option 2
busybox → option

① Simple Init process (Inittab)                                    init2

② Configuration excution faster

folder /etc

① inittab (Configuration that will be parsed by /sbin/init

② Create 3 folders, each folder represent specific runlevels application script (soft link)

③ Create Generic folder that Contains all application script

the runlevels standard name folder

rcx.d

Generic folder

init.d

SystemV provided a Command to start application & Stop the application

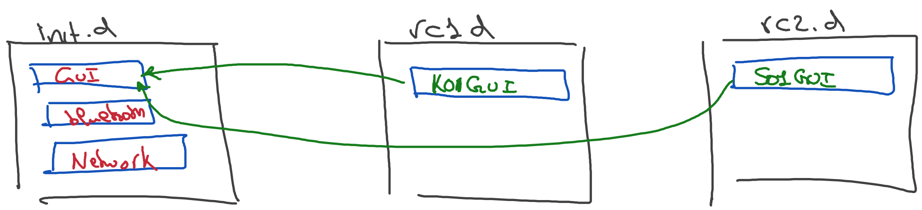
Init.d

① Create script to start & stop the aplication

it will be a bash or it can be another script language

$1

Script stop
Script start




/sbin/init 2

→ parse directory

if ( K )
for ( printf )

if ( S )
for ( pin )

K01syhello Stop

S01 start

Ubuntu

Systemd → dependencies

start → servi

systemV

rc2.d

ssh start